

CAPSTONE PROJECT

End-to-End Sales Data Analysis, Predictive Modeling & Business Intelligence

Prepared By

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Project: End-to-End Analysis Case Study

Tools Used: Python, Pandas, Matplotlib, Seaborn, Scikit-learn, Excel, Power BI

1. Project Overview

The objective of this capstone project is to perform an end-to-end data analysis on a sales dataset and convert the analysis into meaningful business insights.

The project follows the complete data analytics workflow:

Raw Dataset → Data Cleaning → Exploratory Data Analysis → Business Analysis → Regression Modeling → Excel Analysis → Power BI Dashboard → Findings & Recommendations

The project focuses on understanding:

- Sales performance
- Profitability
- Product performance
- Country-wise sales
- Customer segment performance
- Year-wise and monthly sales trends
- Discount analysis
- Sales prediction using regression
- Business recommendations based on the analysis

The dataset used in the project contains **700 rows and 16 columns**.

2. Project Objectives

The main objectives of this project were:

1. Perform **Exploratory Data Analysis (EDA)** on the sales dataset.
 2. Clean and prepare the dataset for analysis.
 3. Identify important sales and profitability patterns.
 4. Analyze sales by country, product, segment and year.
 5. Analyze monthly sales trends.
 6. Analyze profit performance.
 7. Build a **regression model to predict sales**.
 8. Compare Linear Regression and Random Forest Regression.
 9. Export analytical results into Excel.
 10. Build an interactive Power BI dashboard.
 11. Present key findings and business recommendations.
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3. Dataset Description

The dataset contains **700 records and 16 columns**.

The columns are:

Column	Description
Segment	Customer/business segment
Country	Country where the sale occurred
Product	Product sold
Discount Band	Discount category
Units Sold	Number of units sold
Manufacturing Price	Manufacturing cost/price
Sale Price	Selling price
Gross Sales	Sales before discounts
Discounts	Discount amount
Sales	Final sales value
COGS	Cost of Goods Sold
Profit	Profit generated
Date	Transaction date
Month Number	Month number

Column	Description
Month Name	Month name
Year	Year of transaction

The notebook shows that all columns except **Discount Band** initially had complete values. Discount Band had **53 missing values**.

4. Data Loading

The dataset was loaded using Pandas:

```
import pandas as pd

df = pd.read_excel('Project4-dataset.xlsx')
```

The first five rows were inspected to understand the structure and values of the dataset.

The dataset dimensions were then checked:

```
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
```

The result was:

Rows: 700

Columns: 16

5. Data Understanding

The `df.info()` function was used to inspect:

- Column names
- Number of non-null values
- Data types
- Dataset size

The dataset contains:

- Numerical variables such as Sales, Profit, Units Sold and COGS
- Categorical variables such as Country, Product and Segment
- Date-related variables such as Date, Month Number, Month Name and Year

This step helped determine which variables could be used for numerical analysis, visualization and machine learning.

6. Missing Value Analysis

Missing values were checked using:

```
df.isnull().sum()
```

The analysis showed:

Discount Band → 53 missing values

All other columns had zero missing values.

Handling Missing Values

The missing values in `Discount Band` were replaced with:

```
df["Discount Band"] = df["Discount Band"].fillna("Unknown")
```

After this step, the missing-value check showed **0 missing values across all columns**.

This ensured that the dataset was complete before performing further analysis.

7. Duplicate Value Analysis

Duplicate records were checked using:

```
df.duplicated().sum()
```

The result was:

Duplicate rows: 0

Therefore, no duplicate records needed to be removed.

8. Column Name Cleaning

The notebook also cleaned column names by removing unnecessary spaces:

```
df.columns = df.columns.str.strip()
```

This ensured that column names could be referenced consistently throughout the analysis.

9. Descriptive Statistics

Descriptive statistics were generated using:

```
df.describe()
```

This provided:

- Count
- Mean
- Standard deviation
- Minimum
- 25th percentile
- Median
- 75th percentile
- Maximum

for the numerical variables.

Some important values include:

Metric	Mean	Minimum	Maximum
Units Sold	1,608.29	200	4,492.50
Sale Price	118.43	7	350
Sales	169,609.10	1,655.08	1,159,200
Profit	24,133.86	-40,617.50	262,200

The dataset therefore contains both profitable and loss-making records, as indicated by the negative minimum profit value.

10. Overall Business Performance

The following overall metrics were calculated:

```
print("Total Sales:", df["Sales"].sum())
print("Total Profit:", df["Profit"].sum())
print("Total Units Sold:", df["Units Sold"].sum())
```

Results:

Total Sales

\$118,726,350.26

Total Profit

\$16,893,702.26

Total Units Sold

1,125,806 units

These figures provide an overall picture of the business performance in the dataset.

11. Sales Analysis by Country

Sales were grouped by country using Pandas:

```
sales_country = (  
    df.groupby("Country")["Sales"]  
        .sum()  
        .sort_values(ascending=False)  
)
```

The results were:

Country	Sales
United States of America	\$25.03M
Canada	\$24.89M
France	\$24.35M
Germany	\$23.51M
Mexico	\$20.95M

The **United States of America** had the highest sales among the countries analyzed.

Business Interpretation

The U.S. represents the strongest market in this dataset.

However, the sales difference between countries is not extremely large, which suggests that the business has relatively broad geographical sales distribution.

12. Sales Analysis by Product

Sales were grouped by product.

The results were:

Product	Sales
Paseo	\$33.01M

Product	Sales
VTT	\$20.51M
Velo	\$18.25M
Amarilla	\$17.75M
Montana	\$15.39M
Carretera	\$13.82M

Paseo was the highest-selling product.

Business Interpretation

Paseo generated significantly more sales than the other products.

This makes Paseo an important product for:

- Inventory planning
- Marketing
- Sales strategy
- Revenue forecasting

13. Sales Analysis by Segment

Sales were grouped according to customer segment.

Results:

Segment	Sales	Share
Government	\$52.50M	44.22%
Small Business	\$42.43M	35.74%
Enterprise	\$19.61M	16.52%
Midmarket	\$2.38M	2.01%
Channel Partners	\$1.80M	1.52%

The Government segment contributed the largest share of sales at **44.22%**, followed by Small Business at **35.74%**.

Business Interpretation

The Government and Small Business segments together account for approximately **80% of total sales**.

Therefore, these segments are particularly important for the company's revenue generation.

14. Year-wise Sales Analysis

Sales were grouped by year.

Year	Sales
2013	\$26.42M
2014	\$92.31M

The dataset shows a very large increase in sales from 2013 to 2014.

The notebook calculated the growth rate as:

249.46%

Business Interpretation

The increase indicates significant sales expansion between the two years represented in the dataset.

Because the dataset covers only part of 2013 and then the full year 2014, the year-to-year comparison should be interpreted with this time coverage in mind.

The notebook shows:

- 2013 data: September–December
- 2014 data: January–December

This is an important limitation to mention rather than treating the 249.46% increase as a perfectly like-for-like annual comparison.

15. Monthly Sales Analysis

Monthly sales were calculated using Year and Month Number.

The analysis showed the following 2014 monthly sales:

Month	Sales
January	\$6.61M
February	\$7.30M
March	\$5.59M
April	\$6.96M
May	\$6.21M
June	\$9.52M
July	\$8.10M
August	\$5.86M
September	\$6.40M

Month	Sales
October	\$12.38M
November	\$5.38M
December	\$12.00M

The strongest 2014 month was **October**, followed by **December**.

Business Interpretation

The monthly analysis suggests that sales were not evenly distributed throughout the year.

The peaks in October and December could be investigated further to understand whether they are associated with:

- Product demand
- Customer purchasing cycles
- Seasonal effects
- Promotional activity
- Discounting

16. Profit Margin Analysis

Overall profit margin was calculated using:

```
profit_margin = df["Profit"].sum() / df["Sales"].sum() * 100
```

Result:

Overall Profit Margin = 14.23%

This means that, across the dataset, total profit represents approximately 14.23% of total sales.

17. Exploratory Visualizations

The notebook includes multiple visualizations to make the analysis easier to interpret.

These include:

- Total Sales by Country
- Total Sales by Product
- Sales by Segment
- Sales Trend by Year

- Monthly Sales Analysis
- Profit by Product
- Profit Margin by Country
- Profit Margin by Segment
- Discount Analysis

For example, the country and product analyses were visualized using bar charts to compare sales across categories.

These visualizations form the analytical foundation for the Power BI dashboard.

18. Regression Analysis

The second major objective of the project was to build a regression model to predict sales.

Target Variable

The target variable was:

Sales

The model used information such as:

- Country
- Product
- Segment
- Discount Band
- Other selected numerical variables

Categorical variables were transformed using **OneHotEncoder** and processed using a **ColumnTransformer** before model training.

19. Train-Test Split

The dataset was divided into:

- Training data
- Testing data

The training data was used to build the regression model.

The test data was then used to evaluate how well the model predicted unseen sales values.

20. Linear Regression Model

The first regression model used was:

Linear Regression

The model was trained using a preprocessing pipeline containing categorical encoding and the regression algorithm.

The evaluation results were:

Metric	Result
MAE	\$82,876.54
RMSE	\$114,244.40
R ²	0.7643

Interpretation

The Linear Regression model explains approximately **76.43% of the variance in the test-set sales values**, according to the reported R².

21. Random Forest Regression

The notebook also implemented a second model:

Random Forest Regression

The results were:

Metric	Result
MAE	\$11,129.46
RMSE	\$29,469.93
R ²	0.9843

22. Model Comparison

The notebook directly compared both models:

Model	MAE	RMSE	R ²
Linear Regression	\$82,876.54	\$114,244.40	0.7643

Model	MAE	RMSE	R ²
Random Forest	\$11,129.46	\$29,469.93	0.9843

Best Performing Model

Based on the reported test metrics:

Random Forest performed better than Linear Regression.

It had:

- Much lower MAE
- Much lower RMSE
- Higher R²

Therefore, Random Forest was the stronger model among the two tested approaches for this dataset.

23. Actual vs Predicted Sales

The notebook also generated an **Actual vs Predicted Sales** visualization for the Random Forest model.

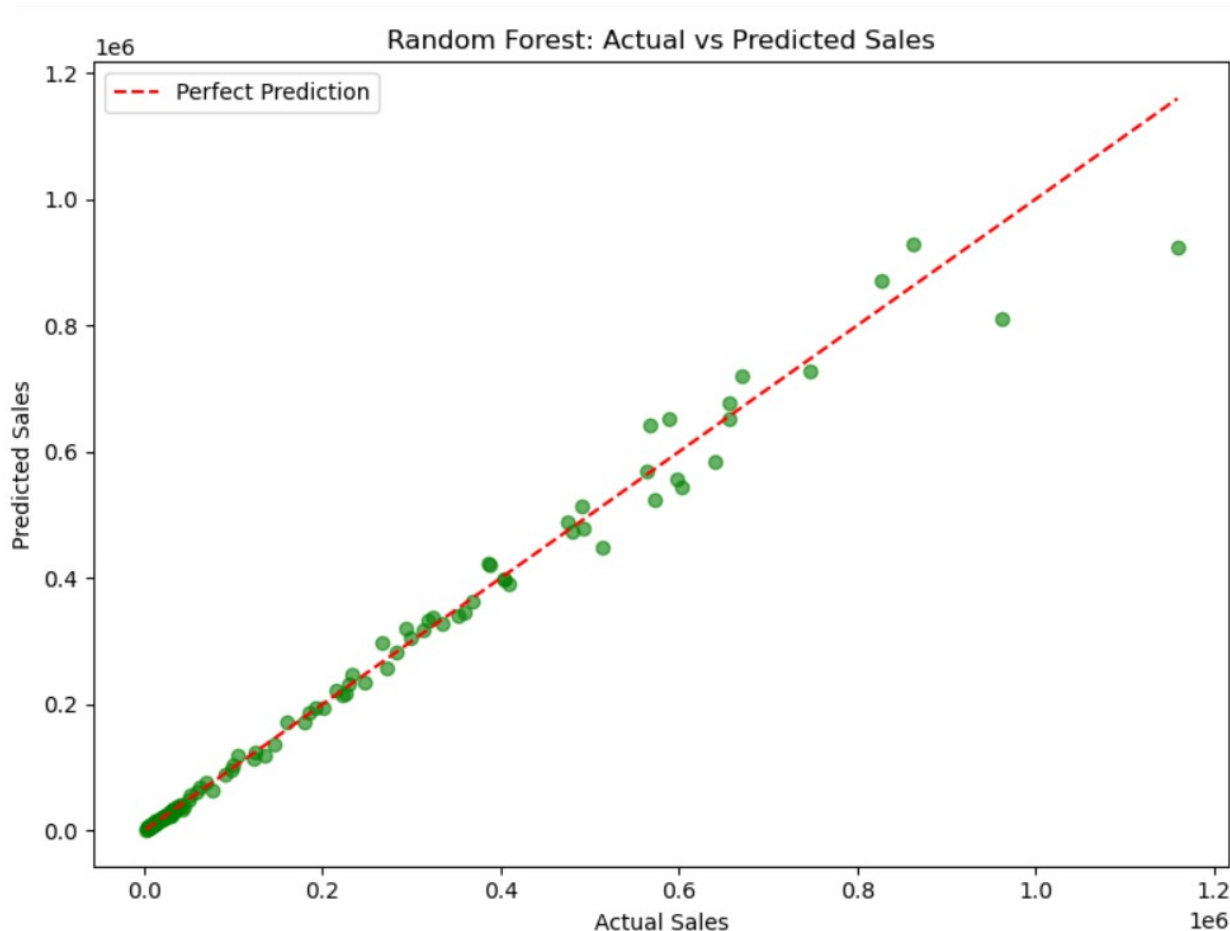
The purpose of this visualization is to compare:

Actual Sales → Predicted Sales

A model with stronger predictions should generally have points closer to the ideal/prediction reference line.

The notebook explicitly labels this visualization:

“Random Forest: Actual vs Predicted Sales.”



24. Important Note About the Regression Result

The Random Forest R^2 value is:

0.9843

This should **not** be described as:

“The model predicts sales with 98.43% accuracy.”

That would be incorrect.

Instead, the correct explanation is:

“The Random Forest model achieved an R^2 score of 0.9843 on the reported test set, indicating a strong relationship between the model's predictions and the observed sales values.”

This is more technically accurate.

25. Excel Analysis

The notebook also generated an Excel workbook called:

Sales_Analysis.xlsx

The notebook uses `pd.ExcelWriter` with the `OpenPyXL` engine to export the analysis.

The workbook contains analytical sheets including:

1. Cleaned Data
2. Sales by Country
3. Sales by Product
4. Sales by Segment
5. Profit by Product
6. Profit by Country
7. Profit by Segment
8. Discount Analysis

This Excel file provides a structured version of the analysis that can be shared with stakeholders or used as a supporting project deliverable.

Sales_Analysis.xlsx - LibreOffice Calc																			
File Edit View Insert Format Styles Sheet Data Tools Window Help																			
Calibri 11 pt B I U A % 00 00 00																			
A1 Segment																			
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
85	Government	United States	Amarilla	High	270	260	350	94500	11340	83160	70200	12960	2014-02-01 00:00:00	2 February	2014				
86	Government	Canada	Carretera	High	923	3	350	323050	41996.5	281053.5	239980	41073.5	2014-03-01 00:00:00	3 March	2014				
87	Government	France	Carretera	High	1790	3	350	626500	81445	545055	465400	79655	2014-03-01 00:00:00	3 March	2014				
88	Government	United States	Montana	High	982.5	5	350	343875	44703.75	299171.3	255450	43721.25	2014-01-01 00:00:00	1 January	2014				
89	Government	Canada	Paseo	High	2632	10	350	921200	119756	801444	684520	117124	2014-06-01 00:00:00	6 June	2014				
90	Government	Germany	Paseo	High	357	10	350	124950	16243.5	108706.5	92820	15886.5	2014-11-01 00:00:00	11 November	2014				
91	Government	Canada	Yelo	High	2632	120	350	921200	119756	801444	684520	117124	2014-06-01 00:00:00	6 June	2014				
92	Government	Canada	Montana	High	2227.5	5	350	779625	109147.5	670477.5	579150	91327.5	2014-01-01 00:00:00	1 January	2014				
93	Government	Germany	Montana	High	1199	5	350	419650	58751	360899	311740	49159	2014-04-01 00:00:00	4 April	2014				
94	Government	Canada	Montana	High	200	5	350	70000	9800	60200	52000	8200	2014-05-01 00:00:00	5 May	2014				
95	Government	Canada	Paseo	High	700	10	350	245000	34300	210700	182000	28700	2014-11-01 00:00:00	11 November	2014				
96	Government	United States	Paseo	High	1177	10	350	411950	57673	354277	306020	48257	2014-11-01 00:00:00	11 November	2014				
97	Government	France	Paseo	High	1922	10	350	672700	94178	578522	499720	78802	2013-11-01 00:00:00	11 November	2013				
98	Government	France	VTT	High	1281	250	350	448350	62769	385581	333060	52521	2013-12-01 00:00:00	12 December	2013				
99	Government	Mexico	Paseo	High	2851	10	350	997850	149677.5	848172.5	741260	106912.5	2014-05-01 00:00:00	5 May	2014				
100	Government	United States	Paseo	High	2007	10	350	702450	105367.5	597082.5	521820	75262.5	2013-11-01 00:00:00	11 November	2013				
101	Government	Mexico	Paseo	High	2151	10	350	752850	112927.5	639922.5	559260	80662.5	2013-11-01 00:00:00	11 November	2013				
102	Government	Canada	Carretera	Unknown	1618.5	3	20	32370	0	32370	16185	16185	2014-01-01 00:00:00	1 January	2014				
103	Government	Germany	Carretera	Unknown	1321	3	20	26420	0	26420	13210	13210	2014-01-01 00:00:00	1 January	2014				
104	Government	France	Montana	Unknown	1899	5	20	37980	0	37980	18990	18990	2014-06-01 00:00:00	6 June	2014				
105	Government	Canada	Paseo	Unknown	292	10	20	5840	0	5840	2920	2920	2014-02-01 00:00:00	2 February	2014				
106	Government	Canada	Paseo	Unknown	1817	10	20	36340	0	36340	18170	18170	2014-12-01 00:00:00	12 December	2014				
107	Government	Canada	VTT	Unknown	1817	250	20	36340	0	36340	18170	18170	2014-12-01 00:00:00	12 December	2014				
108	Government	France	Amarilla	Unknown	1899	260	20	37980	0	37980	18990	18990	2014-06-01 00:00:00	6 June	2014				
109	Government	France	Montana	Low	1375.5	5	20	27510	275.1	27234.9	13755	13479.9	2014-07-01 00:00:00	7 July	2014				
110	Government	France	Yelo	Low	3864	120	20	77280	772.8	76507.2	38640	37867.2	2014-04-01 00:00:00	4 April	2014				
111	Government	United States	Montana	Low	1566	5	20	31320	626.4	30693.6	15660	15033.6	2014-10-01 00:00:00	10 October	2014				
112	Government	United States	Yelo	Low	1566	120	20	31320	626.4	30693.6	15660	15033.6	2014-10-01 00:00:00	10 October	2014				
113	Government	Mexico	Yelo	Low	544	120	20	10880	217.6	10662.4	5440	5222.4	2013-12-01 00:00:00	12 December	2013				
114	Government	United States	Amarilla	Low	1236	260	20	24720	494.4	24225.6	12360	11865.6	2014-11-01 00:00:00	11 November	2014				
Cleaned Data Sales by Country Sales by Product Sales by Segment Profit by Product Profit by Country Profit by Segment Discount Analysis																			
5	Small Business	42427918.5																	
6	Government	52504260.67																	
7																			
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Product	Profit
Paseo	4797437.95
VTT	3034608.02
Amarilla	2814104.06
Velo	2305992.465
Montana	2114754.88
Carretera	1826804.885

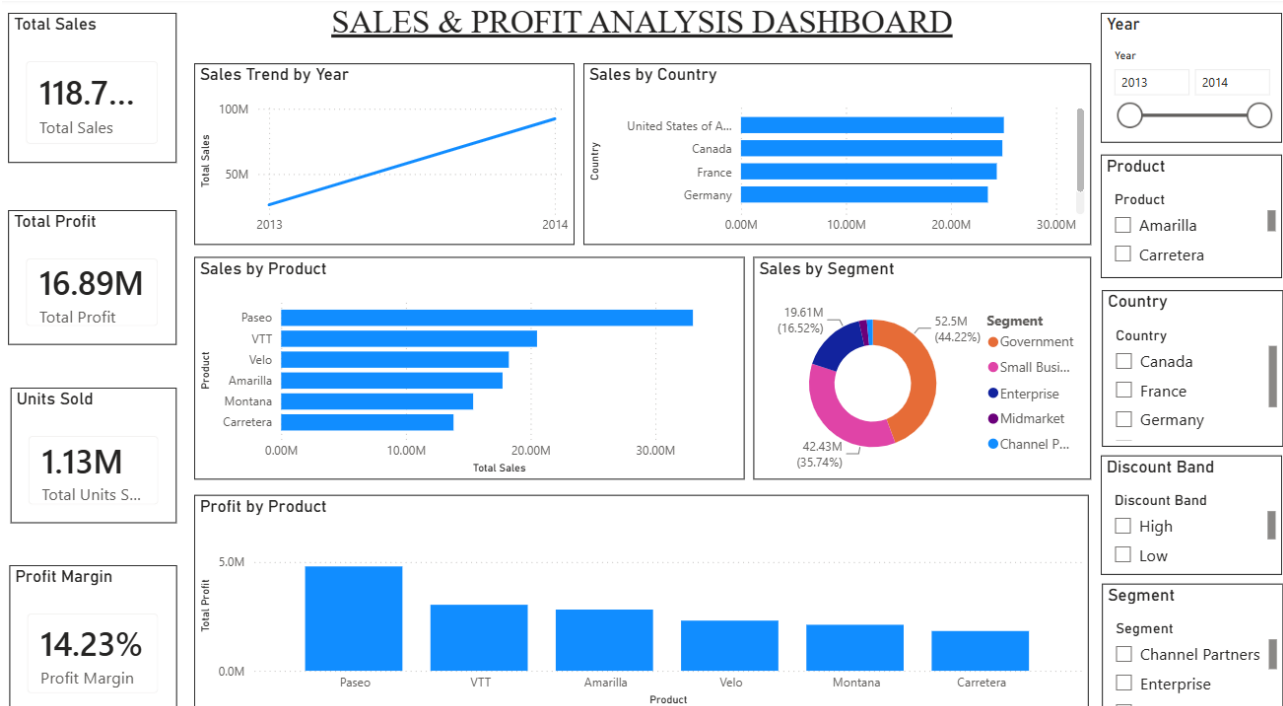
26. Power BI Dashboard

The analysis was then converted into an interactive Power BI dashboard.

The main data source used for Power BI was:

Cleaned Data

The dashboard was designed to provide a stakeholder-friendly overview of the business.



27. Power BI KPI Cards

The dashboard contains four main KPI cards:

Total Sales

\$118.73M

Total Profit

\$16.89M

Total Units Sold

1.13M

Profit Margin

14.23%

These KPIs provide an immediate summary of overall business performance.

28. Power BI Sales Trend

A line chart was created:

Sales Trend by Year

It compares total sales for:

- 2013
- 2014

This allows stakeholders to quickly identify the significant increase in sales between the two years.

29. Power BI Sales by Country

A bar chart was created:

Sales by Country

This visualization allows stakeholders to compare sales across:

- United States of America

- Canada
- France
- Germany
- Mexico

The United States is the highest-selling country in the analysis.

30. Power BI Sales by Product

A bar chart was created:

Sales by Product

This visualization compares all products.

The analysis identifies:

Paseo as the top-selling product.

This makes it easier for stakeholders to identify products that contribute most to revenue.

31. Power BI Sales by Segment

A donut chart was created:

Sales by Segment

The dashboard displays the contribution of each segment.

The largest segment is:

Government — 44.22%

followed by:

Small Business — 35.74%

This visualization helps stakeholders understand where the majority of revenue comes from.

32. Power BI Profit by Product

A profit visualization was added to compare product profitability.

This is important because a product with high sales does not automatically mean it has the highest profit.

Therefore, analyzing both:

Sales + Profit

provides a better understanding of business performance.

33. Power BI Interactive Slicers

Five interactive slicers were added:

1. **Year**
2. **Country**
3. **Product**
4. **Segment**
5. **Discount Band**

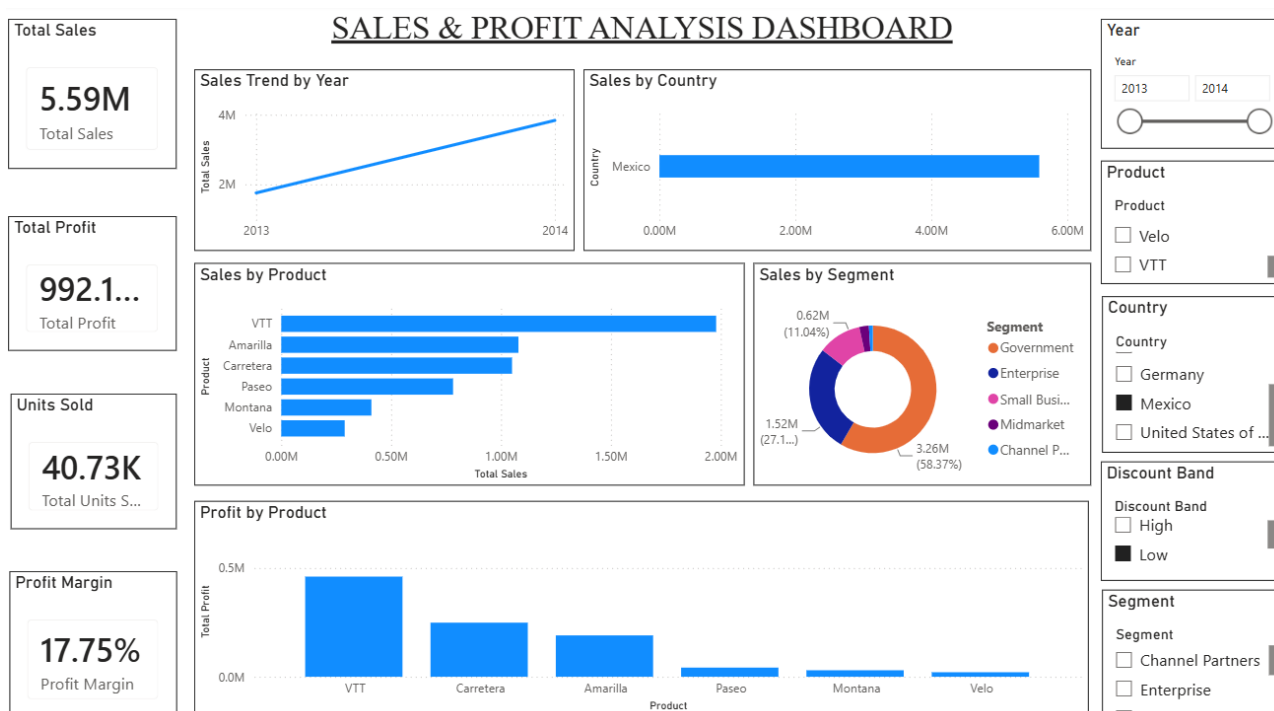
These allow users to filter the dashboard dynamically.

For example, a stakeholder can select:

Year → 2014

and analyze the corresponding sales and profit performance.

Similarly, selecting a specific country or product allows the stakeholder to investigate that particular portion of the business.



34. Key Business Findings

Based on the completed analysis, the major findings are:

Finding 1 — Strong Sales Growth

Sales increased from approximately **\$26.42M in 2013 to \$92.31M in 2014**.

The notebook reports a growth rate of **249.46%**.

Finding 2 — United States is the Leading Country

The United States generated approximately:

\$25.03M in sales

making it the highest-performing country.

Finding 3 — Paseo is the Top Product

Paseo generated approximately:

\$33.01M in sales

making it the highest-selling product.

Finding 4 — Government is the Largest Segment

Government contributed:

44.22% of total sales

while Small Business contributed:

35.74%.

Finding 5 — Profitability

The overall profit margin is:

14.23%

with total profit of approximately:

\$16.89M.

Finding 6 — Random Forest Performs Better

Random Forest achieved:

$R^2 = 0.9843$

compared with:

$R^2 = 0.7643$

for Linear Regression.

It also produced substantially lower MAE and RMSE.

35. Business Recommendations

Based on these findings, the following recommendations can be made.

1. Strengthen High-Performing Markets

The United States is the leading market.

The company should continue supporting strong-performing markets while identifying strategies to increase sales in comparatively lower-performing markets.

2. Focus on High-Performing Products

Paseo is the highest-selling product.

The business should monitor its inventory, demand and marketing performance to maintain its contribution to overall sales.

At the same time, lower-performing products should be analyzed to understand whether the issue is related to pricing, demand, discounts or customer preference.

3. Prioritize Major Customer Segments

Government and Small Business contribute the largest portions of sales.

Therefore, these segments should remain important targets for sales and customer-retention strategies.

4. Monitor Profit Alongside Sales

Sales growth should not be the only performance measure.

The business should monitor:

- Sales
- Profit
- Profit Margin
- COGS
- Discounts

together.

This can help ensure that increasing revenue also produces sustainable profitability.

5. Investigate Monthly Sales Peaks

October and December showed particularly high sales in 2014.

The company should investigate the reasons behind these peaks and determine whether they are associated with seasonal demand, promotions, customer purchasing cycles or other factors.

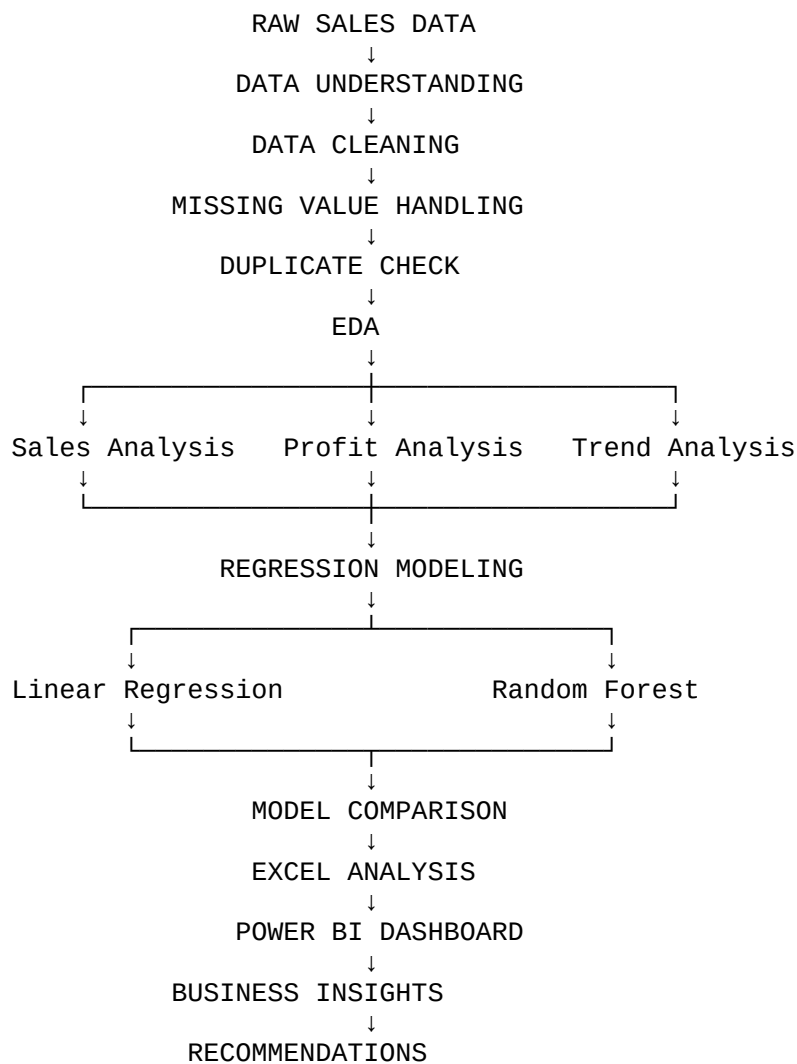
6. Use Predictive Analytics

The Random Forest model showed stronger performance than Linear Regression on the reported test results.

Therefore, Random Forest can be considered as a starting point for sales prediction and planning, with additional validation on future/new data before using it for operational decisions.

36. Project Workflow

The complete workflow of the project can be summarized as:



37. Tools & Technologies Used

Python

Used for:

- Data loading

- Data cleaning
- EDA
- Aggregation
- Statistical analysis
- Visualization
- Regression modeling

Pandas

Used for:

- Data manipulation
- GroupBy analysis
- Missing-value handling
- Aggregation
- Excel export

Matplotlib & Seaborn

Used for:

- Bar charts
- Trend visualizations
- Analytical plots

Scikit-learn

Used for:

- Train-test splitting
- One-hot encoding
- ColumnTransformer
- Linear Regression
- Random Forest Regression
- Model evaluation

Excel

Used to store:

- Cleaned dataset

- Country analysis
- Product analysis
- Segment analysis
- Profit analysis
- Discount analysis

Power BI

Used for:

- KPI cards
 - Sales trend
 - Country analysis
 - Product analysis
 - Segment analysis
 - Profit analysis
 - Interactive slicers
-

38. Deliverables

The completed project contains the following deliverables:

1. Complete Code

`Project4_Capstone.ipynb`

Contains the Python data analysis, visualizations and regression modeling.

2. Excel Analysis

`Sales_Analysis.xlsx`

Contains the cleaned data and analytical summaries.

3. Power BI Dashboard

Interactive dashboard containing KPIs, charts and slicers.

4. PDF Explanation Report

This report documents:

- Project objective

- Dataset
 - Data cleaning
 - EDA
 - Visualizations
 - Regression
 - Model comparison
 - Excel analysis
 - Power BI dashboard
 - Findings
 - Recommendations
 - Conclusion
-

39. Conclusion

This capstone project demonstrates a complete end-to-end data analytics workflow.

The project began with data loading and cleaning, followed by exploratory analysis of sales, products, countries, segments, profitability and trends.

The analysis identified several important business insights, including the strong contribution of the United States, the high sales performance of Paseo, the dominance of the Government segment and significant growth between the two years represented in the dataset.

Two regression approaches were evaluated. Random Forest performed substantially better than Linear Regression according to the reported MAE, RMSE and R^2 metrics.

Finally, the analysis was transformed into Excel outputs and an interactive Power BI dashboard, allowing stakeholders to explore the results through KPIs, charts and filters.

Overall, the project demonstrates how raw business data can be transformed into **actionable insights, predictive analysis and interactive business intelligence**.
